

# CoBit on Verifiable Tasks: a NeurIPS workshop plan

Minimum path to submission — deadline Sat 29 Aug (5 days)

Mon 24 Aug 2026

## 1 The claim, in one paragraph

Every published comparison on Sudoku and TinyGSM→GSM8K — ours included — reports *one sample per problem*. Under that metric the discrete diffusion models MDLM and Duo beat CoBit on GSM8K, but only when they use low-temperature decoding ( $T=0.1$ ), which buys pass@1 by destroying sample diversity. We argue this is the wrong measurement for non-autoregressive reasoning models, and report the first **multi-sample (pass@ $k$  / maj@ $k$ )** evaluation across methods. Our thesis: *CoBit trades in the opposite direction — it has lower pass@1 but substantially higher pass@ $k$  — and pass@ $k$ , not pass@1, is the quantity that bounds what post-training can realise.*

**Why pass@ $k$  is the right proxy.** Yue et al. (NeurIPS 2025, [arXiv:2504.13837](#)) show RLVR mostly *reallocates* probability mass among trajectories already in the base model’s support: it raises pass@1 while often *lowering* pass@ $k$  at large  $k$ . Base-model pass@ $k$  therefore upper-bounds what RL can extract. RL post-training for diffusion LMs exists but is masked-only (d1 / diffu-GRPO, ICLR 2026); continuous diffusion post-training is unexplored. So “how much headroom does a continuous bitstream dLLM have?” is well-posed and unanswered.

**What we will *not* claim.** pass@ $k$  costs  $k \times$  NFE. It is a *headroom* measurement, not a matched-budget win. The paper must say this in the abstract, or a reviewer will say it for us.

## 2 Evidence already in hand (zero new compute)

| Result                         | Value                           | Status                        | Use                          |
|--------------------------------|---------------------------------|-------------------------------|------------------------------|
| Sudoku single-sample (E/M/H)   | 98.5 / 91.7 / 65.9              | <b>HAVE</b> published         | Table 1 anchor               |
| Sudoku churn sweep vs $\gamma$ | Table 16 of COLM ver.           | <b>HAVE</b>                   | appendix                     |
| GSM8K single-sample, 425k      | 29.19% published                | <b>HAVE</b>                   | Table 2 anchor               |
| independent reproduction       | <b>29.42%</b> (388/1319)        | <b>HAVE</b> (this session)    | validates pipeline           |
| GSM8K 500k checkpoint          | 29.34% (full 1319)              | <b>HAVE</b>                   | <i>training is saturated</i> |
| Sudoku-hard maj@16 / pass@16   | $\approx 91\%$ / $\approx 94\%$ | <b>HAVE</b> (indicative $n$ ) | motivates E2                 |
| GSM8K maj@32 / pass@32         | 46.5% / 59.8%                   | <b>HAVE</b> (256-subset)      | motivates E1                 |
| FKC temperature study          | full negative w/ mechanism      | <b>HAVE</b>                   | §4 + appendix                |

The two multi-sample rows are the reason this paper is worth writing: hard-Sudoku maj@16 of  $\approx 91\%$  against a published single-sample 65.9% (Duo: 58.4%), and GSM8K pass@32 of 59.8% against Duo’s best-ever 36%. Both are on reduced sample sets and must be re-run properly — see E1/E2 — but the effect size is not subtle.

## 3 The gap in the literature (verified, not assumed)

I read S-FLM ([arXiv:2605.11125](#)) in full. It is the source of every baseline number in our Table 2/3.

- **No pass@k, no majority vote, no best-of-n, no oracle metric anywhere.** Sudoku: exact match, 180 steps, *one sample per puzzle*. GSM8K:  $T=1$ , 1024 steps, *one generated solution per problem*.
- **Their checkpoints are public** ([github.com/jdeschena/s-flm](https://github.com/jdeschena/s-flm), built on the Duo codebase). *TinyGSM: all baselines released* — AR, MDLM, Duo, FLM, CANDI, S-FLM. Baseline pass@k therefore needs *sampling only, no training*.
- **Sudoku: no checkpoints released**, but training scripts exist and their paper states Sudoku trains in under 2 hours on a single L40S.

## 4 Minimum experiment set (the critical path)

Assume a 4–5 page workshop limit. Nine runs, all sampling, no training.

| ID | Experiment                                     | Config                                      | Runs     | GPU-h                       |
|----|------------------------------------------------|---------------------------------------------|----------|-----------------------------|
| E1 | CoBit GSM8K pass@k                             | $K=16$ , full 1319, 1024 NFE, $\gamma=0.41$ | 1        | 13.5                        |
| E2 | CoBit Sudoku pass@k                            | $K=16$ , full $2k \times 3$ diff., 180 NFE  | 3        | $\sim 6$                    |
| E3 | <b>Duo GSM8K pass@k <math>\times</math> T</b>  | $K=8$ , $T \in \{1.0, 0.1\}$ , 1024 NFE     | 2        | 14                          |
| E4 | <b>MDLM GSM8K pass@k <math>\times</math> T</b> | $K=8$ , $T \in \{1.0, 0.1\}$ , 1024 NFE     | 2        | 14                          |
| E5 | S-FLM GSM8K pass@k                             | $K=8$ , $T=1$ , top-1 velocity              | 1        | 7                           |
|    |                                                |                                             | <b>9</b> | <b><math>\sim 55</math></b> |

**E3/E4 are the paper.** They test the falsifiable prediction: if Duo’s pass@8 *falls* as  $T$  drops to the 0.1 setting that wins it pass@1, the diversity-for-headroom trade is demonstrated rather than asserted. Everything else is context.

**Two technical requirements.** (i) Use the unbiased Codex estimator  $\text{pass}@k = 1 - \binom{n-c}{k} / \binom{n}{k}$ , so one  $K=16$  run yields the *whole* curve  $k=1 \dots 16$  rather than a single point; the matched cross-method region is then  $k \leq 8$ . (ii) All numbers on the *full* 1319 / full 2k. Our first-256 subset runs  $\approx 6$  points hot (35.6% vs 29.4% at identical config) — a subset would silently invalidate every comparison.

## 5 Deliberately dropped

- **DROP Sudoku baselines.** Needs training 4 models  $\times$  3 difficulties. Cite published single-sample numbers instead and report CoBit pass@k alone on Sudoku, stating plainly that no prior work reports it.
- **DROP FLM / CANDI on GSM8K.** They score 0.3% / 0.2% single-sample; pass@k of a model that never solves anything is uninformative.
- **DROP New temperature/FKC compute.** The study is complete; it costs nothing to write up.
- **DROP LM1B, OWT, CoBit-M, CFG,  $k > 16$ .** Different paper.

## 6 HPC logistics (CSD3)

The TinyGSM checkpoints already live on CSD3 at `/rds/project/rds-LlrDsbHU5UM/gb511/projects/Bitstream` — this is where we pulled them from, so E1 needs no data movement. Steps:

1. Push branch `tasks/fkc-temperature` (the GSM8K pass@k eval wiring is currently *uncommitted local work*: `gsm8k_eval.py`, `sandbox_gsm8k.py`, plus the sweep scripts). **This is the one true blocker** — nothing runs on HPC until it is pushed.
2. Clone `s-flm` on CSD3, create its env, pull the TinyGSM checkpoints from HuggingFace.

3. Add a `pass@k` wrapper to their sampler: generate  $K$  samples/problem, keep per-problem correctness vectors, apply the unbiased estimator. Their eval already executes programs in a sandbox, so this is a loop and a JSON field.
4. Submit as array jobs, one cell per (model,  $T$ ); each cell is a few GPU-hours.

## 7 Timeline

| Day            | Work                                                               | Gate                         |
|----------------|--------------------------------------------------------------------|------------------------------|
| Mon 24 (today) | push branch; submit E1 + E2 on CSD3                                | jobs queued                  |
| Tue 25         | s-flm env + checkpoints; <code>pass@k</code> wrapper; submit E3/E4 | <b>go/no-go on baselines</b> |
| Wed 26         | E3/E4 land; E5 if clean; first figures                             | thesis confirmed or flipped  |
| Thu 27         | write: intro, setup, results, FKC section                          | full draft                   |
| Fri 28         | polish, appendix, reproduction checks                              | submission-ready             |
| Sat 29         | buffer + submit                                                    | —                            |

## 8 Paper skeleton

- **§1 Intro** (0.75p). Single-sample accuracy mismeasures non-AR reasoning models; low- $T$  buys `pass@1` with diversity; `pass@k` bounds post-training headroom.
- **§2 Setup** (0.5p). S-FLM parity protocol, unchanged; multi-sample extension; the  $k \times$  NFE caveat stated up front.
- **§3 Results** (1.5p). **Table 1** Sudoku `pass@k/maj@k` vs published single-sample. **Table 2** GSM8K `pass@k` across methods  $\times$  temperature. **Fig 1** `pass@k` vs  $k$  per method. **Fig 2** `pass@8` vs  $T$  for Duo/MDLM (the money plot).
- **§4 Why not sharpening?** (0.5p). The FKC negative, with mechanism — free content.
- **§5 Discussion** (0.25p). Headroom for continuous-dLLM post-training; d1/diffu-GRPO is masked-only.
- **Appendix**. FKC details, protocol, the 29.42% reproduction, saturation at 500k.

## 9 Risks, and what we do about each

| Risk                                                                                       | Mitigation                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Duo’s <code>pass@k</code> does <i>not</i> fall at low <math>T</math> — thesis flips</b> | This is still a publishable finding (“low- $T$ is free: it costs no headroom”), but the paper becomes framing 2, “strongest continuous model on verifiable tasks”, with <code>pass@k</code> as support. Decide Wed, not Fri.      |
| s-flm env / checkpoint mismatch eats a day                                                 | Fall back to CoBit-only <code>pass@k</code> on both tasks + the observation that no prior work reports it. Weaker but still a complete paper. Gate this Tue.                                                                      |
| CSD3 queue times                                                                           | Submit E1/E2 today; use short array cells rather than one long job.                                                                                                                                                               |
| Reviewer: “ <code>pass@k</code> is just more compute”                                      | Address in the abstract; report matched-NFE single-sample alongside; cite Yue et al. for why the quantity matters independently of budget.                                                                                        |
| Config traps ( <code>sigma_data</code> , test subset)                                      | Both bit us this week. <code>sigma_data=0.399844765663147</code> must be passed explicitly (config default 0.5 is wrong and silent); all numbers on full test sets. Re-verify every table row against its JSON before submission. |

## If you only do one thing

Run **E3** (Duo pass@8 at  $T=1.0$  vs  $T=0.1$ ). It is 14 GPU-hours, needs no training, and it alone decides whether the paper has a thesis or a table.